SHEET 3 – **OPTIMIZATION FOR DATA SCIENCE** SS 2026
DATE: 16.03.2026

Exercise 1.1

1. Compute the gradient $\nabla f(x)$ and Hessian $\nabla^2 f(x)$ of the Rosenbrock function
$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2. \tag{1}$$
Show that $x^* = (1, 1)^T$ is the only local minimizer of this function, and that the Hessian matrix at that point is positive definite.

2. Show that the function
$$f(x) = 8x_1 + 12x_2 + x_1^2 - 2x_2^2$$
has only one stationary point, and that it is neither a maximum nor minimum, but a saddle point.

Exercise 1.2

Let $x \in \mathbb{R}^n$, suppose that $f(x) = x^T Q x$, where Q is an $n \times n$ symmetric positive semidefinite matrix. Show that $f(x)$ is convex on the domain $\mathbb{R}^n$ using the convexity defintion.

Exercise 1.3

Let $x, y \in \mathbb{R}$, and $\sigma : \mathbb{R} \to \mathbb{R}$ differentiable function. For parameters $(w, b, v, c) \in \mathbb{R}^4$, define the simple neural network architecture
$$z = wx + b, \qquad a = \sigma(z), \qquad \hat{y} = va + c.$$
The loss function is
$$L : \mathbb{R}^4 \to \mathbb{R}, \qquad L(w, b, v, c) = \tfrac{1}{2}(\hat{y} - y)^2.$$

1. Using the chain rule, compute the partial derivatives
$$\frac{\partial L}{\partial w}, \qquad \frac{\partial L}{\partial b}, \qquad \frac{\partial L}{\partial v}, \qquad \frac{\partial L}{\partial c}.$$

2. Identify quantities that appear repeatedly in several of the partial derivatives. Explain how this observation suggests a way to compute all partial derivatives efficiently.

3. Suppose that the activation function is the identity function, i.e. $\sigma(t) = t$. Is L convex with respect to the parameters (w, b, v, c)?

Exercise 1.4 (Optional)

Prove Second-Order Sufficient Conditions: Suppose that $\nabla^2 f$ is continuous in an open neighborhood of x^* and that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite. Then x^* is a strict local minimizer of f.